

Homework 10

1. In this problem, we will get comfortable with the perturbation theory formalism. Let us set up a simple problem which we can analytically solve, and then recover the solution using Eigenvalue perturbation theory. Consider the two mass problem in 1d with a potential consisting of two parts:

$$V = V_0 + V_1 \quad V_0 = \frac{1}{2}\gamma_a x_a^2 + \frac{1}{2}\gamma_b x_b^2 \quad V_1 = \gamma_o x_1 x_2$$

We can construct the stiffness matrices as usual:

$$\hat{\mathbf{V}} = \hat{\mathbf{V}}_0 + \hat{\mathbf{V}}_1 \quad \hat{\mathbf{V}}_0 = \begin{bmatrix} \gamma_a & 0 \\ 0 & \gamma_b \end{bmatrix} \quad \hat{\mathbf{V}}_1 = \gamma_o \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

We already found the Evals of $\hat{\mathbf{V}}$ in the beginning of the class:

$$\begin{bmatrix} \gamma_a & \gamma_o \\ \gamma_o & \gamma_b \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \lambda \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \quad \rightarrow \quad \lambda = \frac{\gamma_a + \gamma_b}{2} \pm \frac{1}{2} \sqrt{(\gamma_b - \gamma_a)^2 + 4\gamma_o^2}$$

$$\tan \theta = \left(\frac{\gamma_b - \gamma_a}{2\gamma_o} \pm \sqrt{\left(\frac{\gamma_b - \gamma_a}{2\gamma_o} \right)^2 + 1} \right) \quad c_1 = \frac{1}{\sqrt{1 + \tan^2 \theta}} \quad c_2 = \frac{\tan \theta}{\sqrt{1 + \tan^2 \theta}}$$

Let us focus on the minus root (i.e. Eval γ_a for unperturbed stiffness matrix); let us call this λ_- for clarity. Assume $\gamma_b - \gamma_a > 0$.

- (a) Taylor series expand λ_- to second order in γ_o . Also Taylor series expand c_1^- and c_2^- to first order in γ_o .
 - (b) Use Eigenvalue perturbation theory to obtain λ_- to second order in γ_o . Also, use Eigenvalue perturbation theory to obtain $|C_- \rangle$ to first order in γ_o . These results should reproduce what you found in part (a).
2. Graphene is a 2d crystal, and therefore has three independent components of strain: $\epsilon_{xx}, \epsilon_{yy}, \epsilon_{xy}$. Symmetrize these strain components such that they transform like irreducible representations of C_{6v} .
 3. Now that we know the strains in terms of irreducible representations, determine the allowed irreducible derivatives at 2nd and 3rd order.
 4. Reconsider the quadratic potential of graphene with nearest neighbor springs γ_σ and γ_π (in-plane displacements) from HW7.
 - (a) Construct $\hat{\mathbf{D}}_{\mathbf{k}}$ to second order in $\mathbf{k}$. Plot the Eigenvalues along the same k-path as HW7 to confirm that the exact results are produced for small $\mathbf{k}$.
 - (b) Construct the acoustic matrix $\hat{\mathbf{A}}_{\mathbf{k}}$. Again, plot the Eigenvalues along the same k-path to confirm exact acoustic modes are recovered for small $\mathbf{k}$.